

FEASIBILITY STUDY

Fungi Labs Multi-Speciality Hospital

A 100-bed greenfield tertiary-lite facility at Nashik, Maharashtra

Particulars	Detail
Project	Fungi Labs Multi-Speciality Hospital Pvt. Ltd.
Facility	100 operational beds, balanced multi-specialty, 33 critical-care beds
Location	Nashik, Maharashtra — Tier-2, primary catchment approx. 21 lakh
Site	Greenfield: 2.0 acres freehold, 1,05,000 sq ft built-up
Cost of project	Rs 110.78 Cr (Rs 11,078 Lakh)
Proposed funding	70:30 — term loan Rs 77.54 Cr, promoter equity Rs 33.23 Cr
Study horizon	10 operating years, FY2029 to FY2038
Units	Indian Rupees. Rs Lakh unless stated. 100 Lakh = 1 Crore.
Document	Feasibility Study — 1 of 4 in the appraisal set
Date	8 August 2026

Basis of preparation and limitations

Fungi Labs is a fictitious entity and this is an illustrative case study prepared for teaching purposes. It is not an offer, an investment recommendation, or professional advice.

Sector benchmarks for occupancy, ARPOB growth and operating margin are drawn from published ICRA research on the Indian hospital industry. Catchment demography, competitor positions, cost rates, manpower norms and clinical assumptions are reasoned estimates constructed for the case study; they have not been sourced from primary market research, verified against site data, or reviewed by a clinician.

Every figure quoted in this document is generated directly from the accompanying financial model, so the two cannot disagree. That guarantees internal consistency — it does not make the underlying assumptions true. Read Section 9 before relying on any conclusion here.

Contents

If the table of contents appears blank, open the document in Word and press Ctrl+A then F9 to refresh field codes.

1. Executive summary

This study tests whether a 100-bed multi-speciality hospital at Nashik is worth building. The short answer is yes — the asset earns a project IRR of 22.1% against a weighted average cost of capital of 12.5%, and across 10,000 simulated futures it produces a positive net present value 91% of the time. The longer answer is that the project is economically sound and financially fragile, and that the distinction matters more than the headline return.

The economics work because a hospital is a fixed-cost business that becomes very profitable once it is full. Roughly 60% of the cost base does not vary with occupancy. Every additional occupied bed-day therefore drops a large contribution straight to EBITDA. The model shows the operating margin moving from negative in Year 1 to about 24% by Year 5 and 27% by Year 10, entirely on the strength of that leverage.

The fragility comes from the same arithmetic running in reverse. Fixed costs, depreciation and interest do not fall when volumes disappoint. A 20% shortfall in revenue removes considerably more than 20% of EBITDA. In the pessimistic and lender-stress scenarios the project does not gently underperform — it fails to return the equity at all. The simulation puts a 27% probability on the hospital failing to cover its debt service in at least one year.

The four findings that matter

1. The project is viable. Project IRR 22.1% against a 12.5% WACC; equity IRR 26.9%; NPV of Rs 123.19 Cr. Revenue reaches Rs 104.4 Cr by Year 5 and Rs 158.3 Cr by Year 10, against Rs 110.78 Cr invested.
2. It is bankable only if the repayment profile respects the ramp. Deferring the first principal instalment to operating Year 4 — a 36-month moratorium after commissioning — lifts the minimum DSCR from 0.27x to 1.46x and removes every covenant breach. Leverage was never the problem; timing was.
3. The promoter must fund losses the bank will not. A greenfield hospital consumes cash for roughly two years. The model requires Rs 17.80 Cr of promoter cash-flow support on top of the Rs 33.23 Cr equity, and the simulation says a committed standby of about Rs 25 Cr would be needed to cover 85% of outcomes.
4. Demand-side execution dominates everything else. The four largest drivers of equity IRR are all commercial — ramp speed, ARPOB level, ARPOB escalation and peak occupancy. Together they swing the return roughly four times as far as the entire financing package. Interest rate ranks second-last of twelve.

1.1 Recommendation

Proceed to detailed project report and lender appraisal, subject to four conditions:

1. Secure the clinical team before the building. Written commitments from anchor consultants in cardiology, orthopaedics, obstetrics and critical care should be a condition precedent to financial closure, not a task for the commissioning year. The tornado analysis in Section 7 is unambiguous that this is where the value sits.
2. Negotiate the repayment structure, not the interest rate. A 36-month principal moratorium is worth several multiples of what 100 basis points of pricing is worth. Section 8 of the Viability Report quantifies both.
3. Commit the standby facility in writing and size it off the distribution, not the base case. Rs 25 Cr, documented as an unconditional promoter undertaking with a non-withdrawal covenant.
4. Commission independent market research before committing capital. The catchment analysis in Section 3 is constructed from published norms and reasoned estimates. It is adequate for deciding whether to investigate further; it is not adequate for deciding to spend Rs 110 Cr.

2. Scope, approach and what "feasible" means here

A feasibility study answers a narrower question than it is usually asked to. It does not predict what will happen. It establishes whether a proposition is capable of working, identifies the conditions under which it works, and quantifies what happens when those conditions are not met. This study is organised around six tests, each of which can fail independently:

Test	The question it answers	Section	Verdict
Market and demand	Is there enough unmet demand in the catchment to fill 100 beds at the assumed tariff?	3	Conditional pass
Technical and physical	Can the facility be built on this site, to NABH norms, within the space and cost assumed?	4	Pass
Operational and clinical	Can it be staffed and run at the assumed productivity?	5	Conditional pass
Legal and regulatory	Can every licence be obtained, and what is the critical path?	6	Pass
Financial	Does it earn more than its cost of capital, and can it service its debt?	7	Pass, structured
Risk	What is the probability distribution of outcomes, not just the base case?	8	Material risk

"Conditional pass" means the test is met only if a specific, stated condition holds. Those conditions are collected in Section 10.

2.1 Method

Demand is estimated top-down from catchment population and published incidence and hospitalisation rates, then cross-checked bottom-up against the throughput 100 beds can physically deliver. Revenue is built from operating drivers — bed-days multiplied by average revenue per occupied bed-day for inpatients, footfall multiplied by ticket size for outpatients — never as a growth rate applied to a starting number. Costs are separated into variable and fixed so that operating leverage is visible rather than buried. Risk is assessed three ways: discrete scenarios, one-at-a-time sensitivity, and a 10,000-trial Monte Carlo simulation in which every input varies simultaneously.

All four documents in this set, and the workbook, are generated from a single model. There is no second set of numbers.

3. Market and demand feasibility

3.1 Sector context

The Indian hospital sector is in an unusually favourable phase. ICRA upgraded its outlook on the sector to Positive, and its FY2026 research reports sector occupancy holding at 62–64%, average revenue per occupied bed-day growing 6–8% a year, and operating margins stable at 22–24%. Eleven listed operators and two large unlisted players are adding roughly 14,500 beds over FY2026–FY2027 at a capital outlay of about Rs 30,000–32,000 crore, and ICRA still expects debt coverage metrics to remain healthy across that expansion.

Three structural forces sit behind those numbers, and all three favour a tier-2 entrant specifically. Insurance penetration continues to widen, which converts latent demand into paying demand. Government schemes — principally Ayushman Bharat (PMJAY), CGHS and ESI — have made tertiary care accessible to households that previously could not consider it, and tier-2 and tier-3 districts are where that shift is most pronounced. And the supply of organised private capacity remains heavily skewed to metros, so patients from towns such as Nashik have historically travelled to Mumbai or Pune for anything beyond secondary care. Every one of those journeys is addressable demand.

Source for sector benchmarks: ICRA Indian Hospital Industry research, FY2026 and FY2027 outlook publications. Every other figure in this section is a constructed estimate — see 3.5.

3.2 Catchment and addressable demand

The primary catchment is Nashik city and its immediate peri-urban belt, taken at approximately 21 lakh people, with a secondary catchment across the wider district contributing referral volume for tertiary procedures. The demand estimate below is deliberately conservative: it applies published hospitalisation rates to the population, then discounts heavily for the share that will use public facilities or existing private capacity.

Step	Basis	Estimate
Primary catchment population	Nashik city and peri-urban belt	21,00,000
Annual hospitalisation episodes	At approx. 3.4% of population p.a.	71,400
Addressable by organised private care	Approx. 45% — balance to public and informal providers	32,130
of which higher-acuity (secondary / tertiary)	Approx. 55% of the addressable pool	17,672
Admissions the facility needs in Year 5	From the model: occupied bed-days ÷ ALOS of 3.9 days	6,536
Required share of the addressable pool	Model requirement ÷ organised-private-addressable pool	20.3%
equivalent share of the higher-acuity subset only	Shown for reference — the facility also carries general-specialty volume	37.0%

Reading the demand test

The hospital needs roughly 6,536 admissions in Year 5 to reach 70% occupancy. Against an estimated 32,130 annual episodes addressable by organised private care in the primary catchment, that is a required market share of about 20%.

The choice of denominator matters here and is easy to get wrong. Fungi Labs is a full multi-specialty facility — general medicine, general surgery, obstetrics and paediatrics alongside the higher-acuity programmes — so the relevant pool is all organised private admissions, not only the tertiary subset. Measured against the higher-acuity subset alone the figure would read 37%, which overstates the requirement by ignoring the general-specialty volume the hospital also

Step	Basis	Estimate
carries. A required share of roughly a fifth of organised private admissions, built over five years, is achievable for a credible facility with credible clinicians. That is why the demand test passes rather than fails. The estimate also excludes district referral volume and the patients currently travelling to Mumbai and Pune, both of which are genuine upside to it. The condition attached to the pass is that the funnel rests on assumed hospitalisation and addressability rates rather than primary research. If the true addressable pool is half the size assumed here, the required share rises to about 41% — and at that level the conclusion of this section would not survive. Commissioning independent market research is the first recommendation of this study for exactly that reason.		

3.3 Occupancy ramp and the case for its shape

Occupancy is the single most important assumption in the study, and the shape of the ramp matters as much as its destination. New hospitals do not fill linearly. The first year is spent building referral relationships and empanelling with insurers and government schemes; volume accelerates sharply in years two and three as those channels mature; the final approach to stabilised occupancy is slow because it depends on winning complex cases from established competitors.

Demand build	Y1 FY2029	Y2 FY2030	Y3 FY2031	Y4 FY2032	Y5 FY2033	Y6 FY2034	Y7 FY2035	Y8 FY2036	Y9 FY2037	Y10 FY2038
Occupancy	27.0%	43.0%	56.0%	64.0%	70.0%	72.0%	72.0%	72.0%	72.0%	72.0%
Occupied bed-days	9,986	15,768	20,498	23,389	25,491	26,280	26,280	26,280	26,280	26,280
Admissions	2,560	4,043	5,256	5,997	6,536	6,738	6,738	6,738	6,738	6,738
OPD footfall per day	100	160	220	260	290	310	325	335	345	355
Gross ARPOB (Rs/bed-day)	26,500	28,487	30,624	32,920	35,389	38,044	40,897	43,964	47,262	50,806
Net realised ARPOB (Rs)	24,088	25,895	27,837	29,925	32,169	34,582	37,175	39,964	42,961	46,183

Peak occupancy of 72% sits at the upper end of the ICRA sector range of 62–64% but within the 65–75% band that well-run tier-2 facilities achieve. Net ARPOB reaches Rs 32,169 by Year 5, consistent with listed tier-2 operators reporting ARPOB of Rs 30,000–35,000.

3.4 Competitive position

The competitive question in a tier-2 city is not whether rivals exist — they do — but whether they compete for the same case mix. Nashik has established private hospitals and a government medical infrastructure, but organised capacity in the higher-acuity specialties is thin. The strategy that follows from that is deliberately narrow: do not attempt to out-compete incumbents on general medicine and routine surgery, where they have referral incumbency and lower cost structures. Compete where capacity is genuinely absent.

Competitive segment	Incumbent strength	Fungi Labs position
General medicine, routine surgery	Strong — established local players, entrenched referrals	Participate for base occupancy; do not lead with it
Cardiac sciences (cath lab)	Limited local interventional capacity	Primary differentiator and anchor programme
Orthopaedics, joint replacement	Fragmented across single-specialty practices	Strong opportunity — high ARPOB, plannable volume

Competitive segment	Incumbent strength	Fungi Labs position
Obstetrics, neonatal (NICU)	Present, but limited Level-III neonatal capability	Strong opportunity — builds family relationships
Critical care (33 beds)	Constrained across the catchment	Structural advantage; supports referral inflow
Oncology, transplant, neurosurgery	Patients travel to Mumbai and Pune	Out of scope in Phase 1 — revisit at expansion

The cath lab deserves particular comment because it drives a disproportionate share of the case for the project. It is the largest single equipment item at Rs 6.50 Cr, it justifies the coronary care unit, and it is what allows the hospital to describe itself as tertiary-lite rather than as another nursing home. Without an interventional cardiology programme the ARPOB assumptions in this study are not defensible, and the project becomes a materially lower-return proposition.

3.5 What would falsify the demand case

It is worth stating plainly what evidence would overturn the conclusion of this section, because a feasibility study that cannot be falsified is marketing:

- Primary research showing the addressable pool is materially smaller than the $\text{\$}\{ \text{fmt.n}(F.\text{demand.addressable}) \}$ episodes assumed — a halving would take the required market share to about $\text{\$}\{ \text{fmt.pc}(F.\text{demand.required_share_if_half}, 0) \}$, where achievability becomes genuinely doubtful.
- A competing tertiary facility announcing entry into the same catchment before Fungi Labs commissions, which would compress both the ramp and the tariff simultaneously — the combination the pessimistic scenario models.
- Inability to recruit interventional cardiology and orthopaedic anchors, which would remove the differentiated case mix and with it the ARPOB assumption.
- A structural shift in scheme tariffs that pushed the blended realisation index materially below the 0.909 assumed here.

4. Technical and physical feasibility

The facility is planned at 1,05,000 sq ft of built-up area across 2 acres of freehold land — approximately 1,050 sq ft per bed, which is consistent with NABH planning norms for a facility of this acuity and leaves circulation, plant and support space adequate rather than tight. The site allows floor-space-index headroom for a vertical Phase 2, which matters because the cheapest bed a hospital ever builds is the one added to an existing structure.

4.1 Bed configuration

Bed category	Beds	% of total	Planning comment
General ward (multi-bedded)	34	34%	Absorbs scheme and lower-tariff volume
Semi-private (twin sharing)	18	18%	Insurance-segment workhorse
Private / deluxe room	12	12%	Cash-pay segment; highest per-bed tariff
ICU - medical & surgical	14	14%	Highest revenue intensity; drives equipment cost
CCU - coronary care	6	6%	Required by the cath lab programme
NICU / PICU	8	8%	Highest revenue intensity; drives equipment cost
HDU / step-down	5	5%	Step-down; improves ICU throughput
Isolation / negative pressure	3	3%	Post-2020 standard of care
Total	100	100%	33 critical-care beds — 33% of total

The 33% critical-care share is the defining technical choice. It is high for a 100-bed facility and it drives cost in three directions at once: equipment intensity, medical-gas and HVAC engineering, and nursing ratios. It is justified only because critical-care capacity is the scarcest resource in the catchment and because it is what makes the cardiac and neonatal programmes credible. A facility that wanted to minimise capital cost would build 15% critical care and earn a materially lower ARPOB.

4.2 Cost of project

Cost head	Rs Lakh	Rs Cr	% of total
Land (freehold)	1,100	11.00	9.9%
Site development & boundary	150	1.50	1.4%
Civil construction	2,835	28.35	25.6%
MEP & building services	998	9.98	9.0%
Interiors & finishes	546	5.46	4.9%
Medical equipment	2,650	26.50	23.9%
Non-medical equipment & utilities	620	6.20	5.6%
Furniture & fixtures	300	3.00	2.7%
IT, HIS & PACS	185	1.85	1.7%
Pre-operative expenses	450	4.50	4.1%
Contingency	331	3.31	3.0%

Cost head	Rs Lakh	Rs Cr	% of total
Interest during construction (IDC)	563	5.63	5.1%
Margin money for working capital	350	3.50	3.2%
Total cost of project	11,078	110.78	100.0%

Benchmark check	This project	Reference range	Assessment
Cost per bed, all-in	Rs 110.8 L	Rs 30–150 L per bed across listed operators	Within range, mid-band
Cost per bed, excluding land	Rs 99.8 L	Rs 70–100 L per bed for a metro build	Slightly above — see note
Built-up area per bed	1050 sq ft	900–1,200 sq ft per bed, NABH norms	Comfortably within norms
Civil construction rate	Rs 2,700 per sq ft	Rs 2,400–3,200 per sq ft, tier-2	Mid-range

Cost per bed excluding land sits marginally above the metro reference range because this facility carries a cath lab, a CT scanner, four operation theatres and 33 critical-care beds. A 100-bed hospital without interventional cardiology would come in materially lower — and would earn a materially lower ARPOB. The MRI has been deliberately deferred to Year 3 (Rs 550 Lakh) rather than installed at commissioning, because the volumes do not justify it at launch. That is a genuine capital-discipline decision and it is worth about Rs 5.5 Cr of deferred outlay.

4.3 Interest during construction — and why it looks circular

Interest during construction is the one cost head that cannot be estimated independently, because it depends on the size of the loan, which depends on the cost of the project, which includes interest during construction. Most models resolve this by iterating until the number stops moving. That works but it is opaque, and in Excel it usually means enabling iterative calculation, which hides genuine circular-reference errors.

This model solves it algebraically instead. If IDC equals the debt proportion (d) multiplied by the rate (r), the construction period in years (m/12) and the average drawdown factor (f), applied to the total cost, then writing k for the product $d \cdot r \cdot (m/12) \cdot f$ gives $IDC = k \cdot \text{HardCost} \div (1 - k)$. The result is exact, non-circular and auditable in a single cell.

Component	Value	Comment
Hard cost before IDC	Rs 10,515 Lakh	All cost heads plus contingency and margin money
Debt proportion (d)	70.0%	Per the proposed funding structure
Interest rate (r)	10.75%	Mid-point of the 9.5–12.5% market range
Construction period (m)	18 months	Financial closure to commercial operation date
Average drawdown factor (f)	0.45	Average debt outstanding over construction, S-curve drawdown
$k = d \times r \times (m/12) \times f$	0.05079	
Interest during construction	Rs 563 Lakh	$= k \times \text{HardCost} \div (1 - k)$

4.4 Implementation timeline

Phase	Months	Critical path items
Pre-construction	M0–M3	Land conveyance, soil investigation, architectural concept, municipal plan sanction, financial closure
Structure	M3–M10	Foundation, RCC frame, slabs. Weather-exposed — monsoon buffer required
MEP and finishes	M8–M16	HVAC, electrical, medical gas, fire suppression, plumbing. Runs in parallel with structure from M8
Equipment installation	M13–M17	Cath lab and CT require sealed, dust-free, power-stable rooms — the true binding constraint
Licensing and accreditation	M12–M18	AERB for radiology, PCB consent, fire NOC, Clinical Establishments registration, pharmacy and blood-bank licences
Recruitment and trial run	M15–M18	Consultants engaged from M12; nursing and paramedical from M15; two-week trial run
Commercial operation date	M18	Operating Year 1 begins; first revenue recognised

The genuine critical path is not the building. It is the sequencing of equipment installation against licensing against clinician availability. A cath lab cannot be commissioned before AERB clearance; AERB clearance requires the room to be complete; and an interventional cardiologist recruited in month twelve will not wait until month twenty-four for a functioning laboratory. Each month of slippage extends interest during construction and defers the entire revenue ramp — the model shows a nine-month delay costing very little IRR in isolation, which is misleading, because in practice delay arrives alongside cost overrun and a slower ramp.

5. Operational and clinical feasibility

5.1 Manpower plan

Manpower is the largest single cost line and the hardest to reverse. The plan below is built cadre by cadre, from a Year-1 complement of 235 rising to 309 at maturity — approximately 3.09 staff per bed, within the 2.5–3.5 range typical of Indian multi-speciality hospitals.

Cadre	Year 1	Mature	Cost per head (Rs L p.a.)	Comment
Full-time consultants (retainer / employed)	16	20	32.0	The clinical anchors — retention is existential
Resident medical officers / DNB residents	18	24	9.5	Enables 24x7 cover and a DNB teaching programme
Nursing staff	105	140	3.9	Nurse:bed ratio approx. 1.4:1; wage inflation is the binding constraint
Paramedical & technicians	34	46	4.4	
Pharmacy staff	9	12	3.2	
Front office, admissions, billing & TPA	23	31	4.2	Scales with insurance and scheme mix
Finance, HR, IT, quality & NABH	11	15	7.0	
In-house support (biomedical, maintenance)	19	21	2.6	
Total	235	309	—	Year-1 cost Rs 1,494 L

Two design choices in this plan are worth drawing out. First, consultants are split between a full-time salaried core of 16 rising to 20, and a fee-for-service visiting pool paid at 11% of inpatient and outpatient revenue. That hybrid is standard in tier-2 India and it matters financially: the fixed component secures availability and the variable component protects the cost base if volumes disappoint. A hospital that put every consultant on retainer would have a materially worse downside.

Second, the nursing complement is near-complete from commissioning. There is no way to phase it: a licensed 100-bed hospital with 33 critical-care beds must be staffed to run 33 critical-care beds from the day it opens, whether or not they are occupied. This is the single largest contributor to the Year-1 operating loss, and it is unavoidable rather than a planning failure.

5.2 Operating cost structure

Cost line (Rs Lakh)	Y1 FY2029	Y2 FY2030	Y3 FY2031	Y4 FY2032	Y5 FY2033	Y6 FY2034	Y7 FY2035	Y8 FY2036	Y9 FY2037	Y10 FY2038
Consultant payout (fee-for-service)	293	497	697	857	1,005	1,116	1,204	1,297	1,396	1,504
Consumables — drugs, implants, reagents	693	1,176	1,657	2,042	2,401	2,673	2,899	3,126	3,377	3,641
Salaries and wages	1,494	1,857	2,146	2,407	2,646	2,857	3,086	3,333	3,599	3,887
Total operating expenditure	3,400	4,595	5,796	6,852	7,962	8,704	9,378	10,077	10,826	11,617
Total revenue	3,013	5,112	7,205	8,880	10,439	11,622	12,603	13,592	14,684	15,831

Cost line (Rs Lakh)	Y1 FY2029	Y2 FY2030	Y3 FY2031	Y4 FY2032	Y5 FY2033	Y6 FY2034	Y7 FY2035	Y8 FY2036	Y9 FY2037	Y10 FY2038
EBITDA	(386)	518	1,409	2,029	2,478	2,918	3,225	3,515	3,858	4,214
EBITDA margin	-13.0%	10.0%	20.0%	23.0%	24.0%	25.0%	26.0%	26.0%	26.0%	27.0%
Contribution margin ratio	67.0%	67.0%	67.0%	67.0%	67.0%	67.0%	67.0%	67.0%	67.0%	68.0%

Margins reach 24.0% in Year 5 and 27.0% in Year 10, against the ICRA sector benchmark of 22–24%. The contribution margin ratio of approximately 67% means that once fixed costs are covered, roughly two-thirds of every additional rupee of revenue reaches EBITDA. That is the operating leverage that makes the project attractive — and dangerous.

5.3 Productivity assumptions and their credibility

Metric	Year 5	Year 10	Reference	Assessment
Revenue per bed (Rs Lakh p.a.)	104.4	158.3	Rs 90–130 L at maturity	Consistent
Net ARPOB (Rs per bed-day)	32,169	46,183	Rs 30,000–35,000 tier-2	Consistent
Average length of stay (days)	3.9	3.9	3.5–4.5 multi-specialty	Consistent
Occupancy	70.0%	72.0%	62–64% sector; 65–75% good tier-2	Upper end
Staff per bed	3.09	3.09	2.5–3.5	Lean but workable
Consumables as % of revenue	23.0%	23.0%	20–25%	Mid-range

The assumption carrying the most weight is ALOS held flat at 3.9 days for a decade. In reality two forces pull in opposite directions: clinical efficiency and payer pressure shorten stays, while an enriching case mix towards cardiac and orthopaedic work lengthens them. Holding it constant is a defensible simplification, but it is a simplification, and a reader should know that a half-day movement in either direction shifts admissions by roughly 13% without changing bed-days or revenue at all.

6. Legal and regulatory feasibility

No approval in the list below is discretionary or difficult to obtain for a compliant facility. The risk is not refusal; it is sequencing. Several approvals are prerequisites for others, and two of them — AERB clearance and the fire NOC — sit directly on the critical path to commissioning.

Approval	Authority	Stage	Critical path
Land use conversion and building plan sanction	Nashik Municipal Corporation	Pre-construction	Yes — blocks start
Environmental clearance / PCB consent to establish	Maharashtra Pollution Control Board	Pre-construction	Yes
Registration under the Clinical Establishments framework	State health authority	Pre-commissioning	Yes — blocks opening
AERB licence for radiology and cath lab	Atomic Energy Regulatory Board	Pre-commissioning	Yes — blocks cath lab and CT
Fire safety no-objection certificate	State fire services	Pre-commissioning	Yes — blocks opening
Biomedical waste authorisation	Maharashtra Pollution Control Board	Pre-commissioning	Yes
Pharmacy / drug licence (Form 20–21)	State FDA	Pre-commissioning	Yes
Blood storage centre licence	State FDA / NBTC	Pre-commissioning	For surgical programmes
PNDT registration	District appropriate authority	Pre-commissioning	For obstetric ultrasound
Lift, boiler, DG set and electrical clearances	State inspectorates	Pre-commissioning	Yes
PMJAY / CGHS / ESI empanelment	Scheme authorities	Post-commissioning	Revenue-critical, not opening-critical
NABH accreditation (entry level, then full)	Quality Council of India	Year 1–2	Required for insurance tariffs

The empanelment trap

Scheme and insurance empanelment does not block commissioning, which is why it is routinely under-planned — and it is the most common cause of a slow ramp in a new Indian hospital.

PMJAY, CGHS and ESI empanelment each take months and generally require the facility to be operational and, in some cases, accredited before an application can even be made. NABH entry-level accreditation typically requires several months of documented operating history. A hospital that opens without these in place is a cash-pay-only facility for its first two or three quarters — which is precisely the period in which it can least afford reduced volume.

In this model the blended payer mix assumes 25% of revenue from government schemes and 35% from insurance. If empanelment slips two quarters, that 60% of revenue is unavailable exactly when the Year-1 operating loss of Rs 3.86 Cr is being funded. Applications should be prepared during construction and filed on the first day they are admissible.

6.1 Corporate and tax structure

The entity is assumed to be a private limited company electing the concessional regime under section 115BAA of the Income-tax Act — a 22% base rate plus 10% surcharge and 4% cess, giving an effective 25.168%. That election forgoes most exemptions and incentives but removes minimum alternate tax, which materially benefits a project carrying large early-year losses. Business losses carry forward eight assessment years under section 72; unabsorbed depreciation carries forward indefinitely under section 32(2). The model reflects the accumulated losses of the ramp years sheltering taxable profit into Year 6.

Tax treatment is modelled on book profit for simplicity. A full deferred-tax computation would track the timing difference between straight-line book depreciation and written-down-value tax depreciation; the workbook shows that difference on the Depreciation sheet without carrying a formal DTA/DTL schedule. This is a case-study simplification and would need to be corrected for an actual filing.

7. Financial feasibility

7.1 Revenue build

Revenue line (Rs Lakh)	Y1 FY2029	Y2 FY2030	Y3 FY2031	Y4 FY2032	Y5 FY2033	Y6 FY2034	Y7 FY2035	Y8 FY2036	Y9 FY2037	Y10 FY2038
Inpatient (IPD)	2,406	4,083	5,706	6,999	8,201	9,088	9,770	10,503	11,290	12,137
Outpatient (OPD)	256	433	632	791	935	1,060	1,178	1,287	1,405	1,532
Diagnostics — walk-in and referral	156	276	411	526	635	719	815	891	991	1,081
Pharmacy — outpatient	96	169	251	320	385	436	493	538	597	651
Dialysis	60	95	134	162	189	214	232	249	267	286
Other operating income	40	55	70	82	95	105	115	124	133	142
Total revenue	3,013	5,112	7,205	8,880	10,439	11,622	12,603	13,592	14,684	15,831
Total revenue (Rs Cr)	30.1	51.1	72.0	88.8	104.4	116.2	126.0	135.9	146.8	158.3

Inpatient revenue is bed-days multiplied by net realised ARPOB. By convention ARPOB already includes drugs and diagnostics consumed by inpatients, so the diagnostics and pharmacy lines above capture walk-in and referral business only, avoiding double-counting.

7.2 Profitability

Rs Lakh	Y1 FY2029	Y2 FY2030	Y3 FY2031	Y4 FY2032	Y5 FY2033	Y6 FY2034	Y7 FY2035	Y8 FY2036	Y9 FY2037	Y10 FY2038
Revenue	3,013	5,112	7,205	8,880	10,439	11,622	12,603	13,592	14,684	15,831
Operating expenditure	3,400	4,595	5,796	6,852	7,962	8,704	9,378	10,077	10,826	11,617
EBITDA	(386)	518	1,409	2,029	2,478	2,918	3,225	3,515	3,858	4,214
EBITDA margin	-13.0%	10.0%	20.0%	23.0%	24.0%	25.0%	26.0%	26.0%	26.0%	27.0%
Depreciation and amortisation	614	614	655	670	687	618	607	632	660	689
EBIT	(1,000)	(96)	754	1,359	1,791	2,300	2,618	2,883	3,198	3,525
Finance cost	870	891	906	886	825	729	612	479	329	166
Profit before tax	(1,870)	(987)	(152)	473	966	1,571	2,006	2,404	2,870	3,359
Current tax	0	0	0	0	0	0	505	605	722	845
Profit after tax	(1,870)	(987)	(152)	473	966	1,571	1,501	1,799	2,147	2,513

The shape of this table is the shape of every greenfield hospital. EBITDA turns positive in Year 2 and profit after tax in Year 4. Cumulative losses of roughly Rs 30.09 Cr accumulate across the first three years before the business begins compounding. Nothing about that pattern indicates a bad project; everything about it dictates how the project must be financed.

7.3 Break-even and operating leverage

Break-even analysis	Y1 FY2029	Y2 FY2030	Y3 FY2031	Y4 FY2032	Y5 FY2033	Y6 FY2034	Y7 FY2035	Y8 FY2036	Y9 FY2037	Y10 FY2038
Actual occupancy	27.0%	43.0%	56.0%	64.0%	70.0%	72.0%	72.0%	72.0%	72.0%	72.0%

Break-even analysis	Y1 FY2029	Y2 FY2030	Y3 FY2031	Y4 FY2032	Y5 FY2033	Y6 FY2034	Y7 FY2035	Y8 FY2036	Y9 FY2037	Y10 FY2038
Accounting break-even occupancy	33.0%	37.0%	40.0%	42.0%	45.0%	45.0%	45.0%	44.0%	44.0%	44.0%
Cash break-even occupancy	44.0%	48.0%	50.0%	57.0%	61.0%	61.0%	60.0%	59.0%	57.0%	55.0%
Margin of safety (pp)	-17.0%	-5.0%	6.0%	7.0%	9.0%	11.0%	12.0%	13.0%	15.0%	17.0%
Degree of operating leverage	-2.0x	-35.7x	6.4x	4.4x	3.9x	3.4x	3.3x	3.2x	3.1x	3.0x

The most useful number in this study

Cash break-even occupancy in the stabilised years is approximately 61%, against actual occupancy of 72%. That gap — a margin of safety of roughly 11% percentage points — is the buffer between comfort and distress.

In Year 3 the margin is far thinner, and in Years 1 and 2 it is negative: the hospital cannot cover its obligations from operations, by design. The entire financing structure recommended in this appraisal exists to carry the business across that gap.

If a promoter remembers one figure from this document, it should be this one. Occupancy below the high fifties in a stabilised year means the hospital is not generating enough cash to service its debt, regardless of what the profit and loss account says.

7.4 Returns

Measure	Value	Benchmark or comment
Project IRR (unlevered)	22.13%	The return the asset earns, independent of financing
WACC	12.50%	Cost of equity approx. 17.5% at 40% weight; post-tax debt approx. 8.0% at 60%
Spread over WACC	9.63%	Positive spread means value is created
Project NPV at WACC	Rs 12,319 L (Rs 123.19 Cr)	
Equity IRR (levered)	26.91%	The return the shareholder earns after debt service
Promoter hurdle	18.0%	Typical sponsor hurdle for greenfield healthcare
Money multiple over 10 years	12.3x	Includes exit at the assumed EV/EBITDA multiple
Exit enterprise value, Year 10	Rs 547.80 Cr	Year 10 EBITDA of Rs 42.1 Cr at 13.0x
Enterprise value today (DCF midpoint)	Rs 212.05 Cr	Average of Gordon-growth and exit-multiple methods
Value created over cost of project	Rs 101.27 Cr	DCF enterprise value less what it costs to build

The equity IRR exceeds the project IRR because the asset return of 22.1% is above the post-tax cost of debt of approximately 8.0%. Leverage amplifies the spread. It amplifies losses in the same proportion, which is why Section 8 exists.

Terminal value accounts for 72% of the exit-multiple enterprise value and 66% of the Gordon-growth value. Any DCF in which two-thirds of the answer sits beyond the forecast horizon should be read as an assumption about perpetuity dressed as a calculation. The sensitivity grid on exit multiple against WACC in the workbook shows how wide the plausible range is.

8. Risk assessment

8.1 Scenarios

Metric	Base	Optimistic	Pessimistic	Lender stress
Revenue Year 5 (Rs Lakh)	10,439	13,180	7,393	5,755
EBITDA Year 5 (Rs Lakh)	2,478	4,566	204	(1,023)
EBITDA margin Year 5	23.7%	34.6%	2.8%	-17.8%
First EBITDA-positive year	2	1	4	never
Minimum DSCR	1.46x	2.52x	0.04x	-1.04x
Covenant breach years	0	0	7	7
Minimum cash balance (Rs Lakh)	158	1,384	(15,769)	(29,382)
Project IRR	22.1%	32.9%	-18.5%	n.m.
Equity IRR	26.9%	43.2%	n.m.	n.m.

"n.m." means not meaningful — the cash flows never turn sufficiently positive for an internal rate of return to exist. In plain terms, the equity is not returned.

The asymmetry here is the finding. Moving from base to optimistic adds roughly fourteen percentage points of equity IRR. Moving from base to pessimistic does not subtract fourteen — it destroys the investment case entirely. That is operating and financial leverage compounding: fixed costs do not fall with revenue, depreciation is fixed absolutely, and interest is contractual. A hospital at 45% occupancy is not a slightly less profitable hospital; it is an insolvent one.

8.2 Which drivers actually matter

Driver	Low	High	Equity IRR at low	at high	Swing
Occupancy ramp speed	85.0%	112.0%	18.1%	32.4%	14.3%
ARPOB Year 1 (INR/bed-day)	23,000	30,000	19.3%	32.9%	13.5%
ARPOB escalation p.a.	5.0%	9.5%	18.7%	32.1%	13.4%
Peak occupancy	64.0%	80.0%	20.7%	32.0%	11.3%
Exit EV/EBITDA multiple	9	17	23.1%	29.9%	6.8%
Salary escalation p.a.	10.0%	6.0%	23.5%	29.4%	5.9%
Consumables % of revenue	26.0%	20.5%	23.7%	29.3%	5.6%
OPD footfall	80.0%	120.0%	24.0%	29.5%	5.5%
Consultant payout %	13.0%	9.0%	25.1%	28.6%	3.5%
Capex overrun	15.0%	-5.0%	24.9%	27.6%	2.7%

Where to spend management attention

The top four drivers — occupancy ramp speed, ARPOB level, ARPOB escalation and peak occupancy — are all demand-side, and each swings equity IRR by roughly 10 to 13 percentage points. Interest rate swings it by about 1.5

Driver	Low	High	Equity IRR at low	at high	Swing
points and ranks second-last of twelve. The practical implication runs against instinct. Promoters negotiate hardest over loan pricing because it is the most visible, most negotiable number in the deal. Four percentage points of additional occupancy is worth several times more than 300 basis points of interest. The effort belongs in clinician recruitment, referral-network building and empanelment. The caveat: a tornado moves one variable at a time. Real downside arrives as a correlated bundle — a slow ramp and weak pricing and cost inflation together. That is why the Monte Carlo simulation, not this table, is the honest risk measure.					

8.3 Probability, not just possibility

Scenarios tell you what could happen; they say nothing about likelihood. The model was therefore run 10,000 times with every input drawn simultaneously from a distribution, and with occupancy and ARPOB correlated at +0.35 to reflect that stronger demand usually travels with a richer case mix.

Outcome	Probability	Reading
Project NPV greater than zero	91.4%	The project creates value in roughly nine cases out of ten
Equity IRR above 12% (cost of capital)	89.0%	
Equity IRR above 15%	82.8%	
Equity IRR above 18% (promoter hurdle)	74.1%	Clears the sponsor hurdle about three times in four
Minimum DSCR at or above 1.25x (covenant)	50.6%	Barely better than a coin flip
Minimum DSCR at or above 1.10x	64.3%	
Minimum DSCR below 1.00x — technical default	26.7%	Roughly a one-in-four chance
Median equity IRR	23.4%	5th percentile 6.7%; 95th percentile 34.1%
EBITDA margin Year 5 below 20%	32.3%	Against the ICRA sector range of 22–24%

The central conclusion of this study

A 91% probability of positive net present value sitting alongside a 27% probability of technical default is not a contradiction. It is the signature of a good asset that is being financed aggressively.

The asset is sound: the demand exists, the cost structure works, and the returns comfortably exceed the cost of capital across most of the distribution. The financing is fragile: the covenant holds in only about half of simulated outcomes, and the promoter support sized for the base case is inadequate in roughly half of them.

These are separable problems, and the second one is fixable. Section 10 sets out how, and the Viability Report quantifies each mitigation. The recommendation of this study is not "do not build the hospital" — it is "do not finance it as originally proposed".

8.4 Payer mix — the risk that hides in the balance sheet

Metric	Blended (40/35/25)	Government-heavy (30/25/45)	Cash-premium (60/35/05)
Weighted realisation index	0.9090	0.8590	0.9650
Weighted debtor days	48.0	64.2	26.6
Revenue Year 5 (Rs Lakh)	10,439	9,988	10,945
EBITDA Year 5 (Rs Lakh)	2,478	2,180	2,811
EBITDA margin Year 5	23.7%	21.8%	25.7%
Net working capital Year 5	1,016	1,414	426
Minimum DSCR	1.46x	1.21x	1.73x
Equity IRR	26.9%	23.3%	30.9%

Government scheme volume looks like free occupancy and is not. It is paid for twice: once in realisation, because PMJAY and similar schemes settle at roughly 72% of list tariff, and again in working capital, because those receivables run to around 110 days against three days for cash patients. Shifting from the blended mix to a government-heavy mix costs roughly three percentage points of equity IRR and pushes minimum DSCR close to the covenant floor, while adding several crore of permanent working capital that has to be funded by borrowing at 11.5%.

The strategic implication is not to avoid scheme business — in a tier-2 catchment that would leave beds empty and forgo the social licence that comes with it. It is to price it correctly in the plan, to cap it deliberately as a share of revenue, and to build the billing and documentation discipline that gets claims paid in ninety days rather than a hundred and fifty.

9. Limitations — what this study cannot tell you

A feasibility study is only as good as its weakest assumption, and the honest thing to do is name them rather than let a reader discover them later.

9.1 The demand estimate is constructed, not researched

Catchment population, hospitalisation incidence, addressability and the achievable share are reasoned estimates built from published norms. No patient-flow survey, physician interviews, competitor bed-census or payer-mix study underpins them. The cross-check in Section 3.2 — that the required market share is roughly a fifth of the addressable pool rather than a half — is what makes the conclusion robust to moderate error. It would not survive an error of a factor of two: halving the pool takes the requirement to about \$`{fmt.pc(F.demand.required_share_if_half,0)}`. Primary market research is the single highest-value piece of additional work available and should precede any capital commitment.

9.2 The Monte Carlo distributions are judgemental

The input distributions are not fitted to a dataset of comparable greenfield hospitals, because no such public dataset exists at this granularity. They are informed guesses about plausible ranges. Only one correlation is modelled — occupancy with ARPOB — whereas in reality wage inflation, consumable inflation and interest rates also move together, which would fatten the left tail beyond what is shown. And triangular distributions have no tails past their bounds, so genuine catastrophe is entirely absent from these numbers: a clinical scandal, a licence suspension, the simultaneous departure of the three anchor consultants, a pandemic. The simulation should be read as a structured argument about relative risk, not as a forecast.

9.3 Simplifications that would need correcting in practice

- Tax is computed on book profit. A real model would maintain a deferred tax schedule tracking straight-line book depreciation against written-down-value tax depreciation.
- ALOS is held constant at 3.9 days for ten years. Clinical efficiency and payer pressure would shorten it; case-mix enrichment would lengthen it.
- The working-capital facility is drawn at a fixed utilisation percentage rather than sized dynamically against the drawing power a lender would actually compute from stock and receivables.
- No inflation differential is modelled between tariff realisation and cost inflation beyond the separate escalation rates applied; a sustained period in which costs outrun tariffs is not explicitly scenario-tested.
- Phase 2 expansion is not modelled. The site carries FSI headroom and the cheapest incremental bed is one added to an existing structure, so the true option value of this project is understated.
- No competitor response is modelled. A rational incumbent will respond to a new tertiary entrant, most likely on price and on consultant retention.

9.4 What would change the recommendation

This study recommends proceeding to detailed appraisal. That recommendation would reverse on any of the following: primary research showing the addressable pool is roughly half the assumed size; failure to obtain written commitments from anchor consultants in cardiology and orthopaedics; a competing tertiary facility announcing entry into the catchment; or the promoter being unable or unwilling to commit the Rs 25 Cr standby facility that the risk analysis shows is necessary.

10. Conditions for feasibility

The verdicts in Section 2 were qualified as conditional. Those conditions are collected here. Each is specific, each is verifiable, and each should be tracked to closure before capital is committed.

#	Condition	Owner	By when
1	Commission independent primary market research validating catchment size, addressable demand and competitor bed census	Promoter	Before financial closure
2	Obtain written commitments from anchor consultants: interventional cardiology, orthopaedics, obstetrics, critical care	Promoter / Medical Director	Condition precedent to closure
3	Negotiate a 36-month principal moratorium from commissioning, with 28 quarterly instalments thereafter	Promoter / CFO	At sanction
4	Commit a Rs 25 Cr standby facility in writing, with a non-withdrawal undertaking for the tenor of the term loan	Promoter	Condition precedent to closure
5	Negotiate a covenant holiday on total-outside-liabilities to net-worth until Year 5, disclosed up front	CFO	At sanction
6	Prepare PMJAY, CGHS, ESI and insurance empanelment applications during construction; file on first admissible day	CEO	Month 15 onward
7	Appoint the CEO and Medical Director no later than month 12 of construction	Promoter / Board	Month 12
8	Fix the equipment procurement contract with liquidated damages tied to the commissioning date	Projects	Month 6
9	Establish the billing and claims function before commissioning, with a target of 90-day scheme realisation	CFO	Month 16
10	Agree board-level operating triggers: occupancy, ARPOB and cash thresholds that force a defined management response	Board	At closure

10.1 Concluding statement

On the evidence assembled here, a 100-bed multi-speciality hospital at Nashik is feasible. It earns a project IRR of 22.1% against a cost of capital of 12.5%, requires roughly a fifth of the organised private admissions in its catchment, can be built within the space and cost assumed, and can be licensed and staffed on the timeline proposed. Across 10,000 simulated futures it creates value 91% of the time.

It is also, as originally proposed, fragile. The financing structure must be rebuilt around the ramp rather than against it, and the promoter must commit materially more contingent capital than the base case suggests. Those are conditions, not objections. Subject to the ten conditions above, this study recommends proceeding to detailed project report and lender appraisal.

This Feasibility Study is document 1 of 4. It is accompanied by the Detailed Project Report, the Viability and Bankability Report, the Learning Companion, the master financial model workbook, and the investor presentation. All numerical content across the set is generated from a single model; the documents cannot disagree with the workbook.